Development of a new structural variant detection software based on graph clustering machine learning algorithms from long reads

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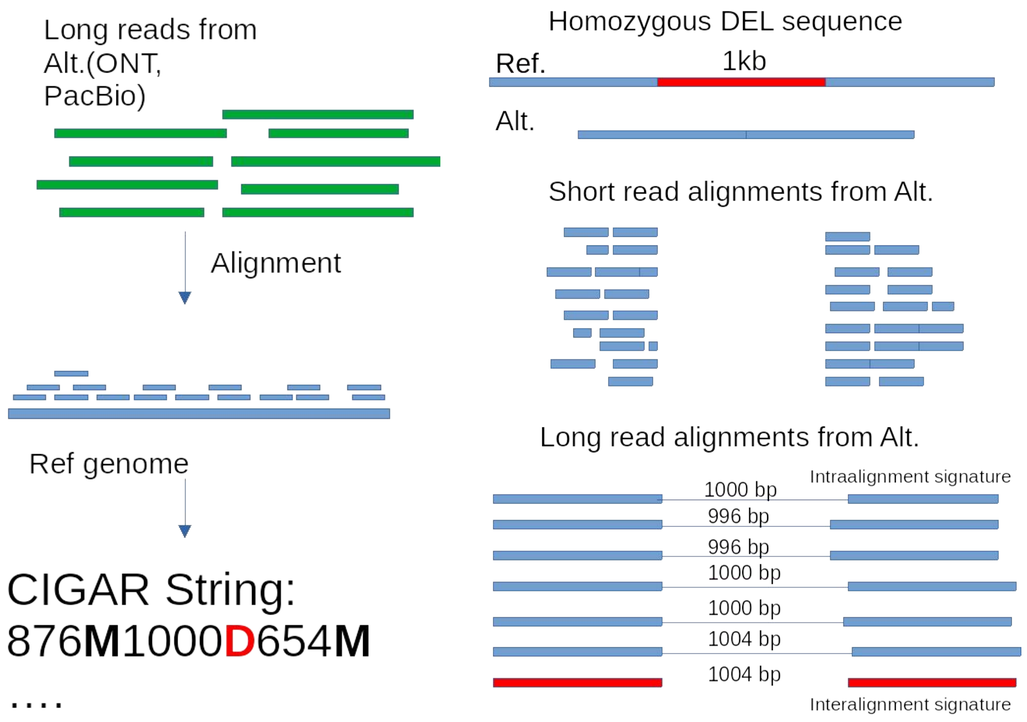
Introduction

Given the importance of structural variants (SV), a large number of computational methods have been developed to identify and genotype SVs, based on high throughput sequencing (HTS) data. Most of these tools are based on short-read sequencing (Cleal et.al., 2022). This presents many limitations, mostly due to the length of structural variants, which usually exceeds the read length. New SV calling tools have adopted long reads as their input data, significantly increasing the accuracy of SV detection (Schwarz et.al., 2021). From this, functionally relevant structural variants have been detected, including some that affect the pathophysiology of diseases such as human cancer (Fujimoto et.al., 2021). However, further improvements could be achieved by novel algorithmic techniques since alignment errors still cause inaccuracies and SV detection tools still lack in terms of certain aspects of software design.

We developed a new software solution that improves the detection of SVs from long read alignments using the DBSCAN algorithm to solve the clustering problem, and that implements a new bayesian genotyping model. Our algorithm is integrated into the bioinformatic software suite (**NGSEP**) to further facilitate the analysis of genomic data.

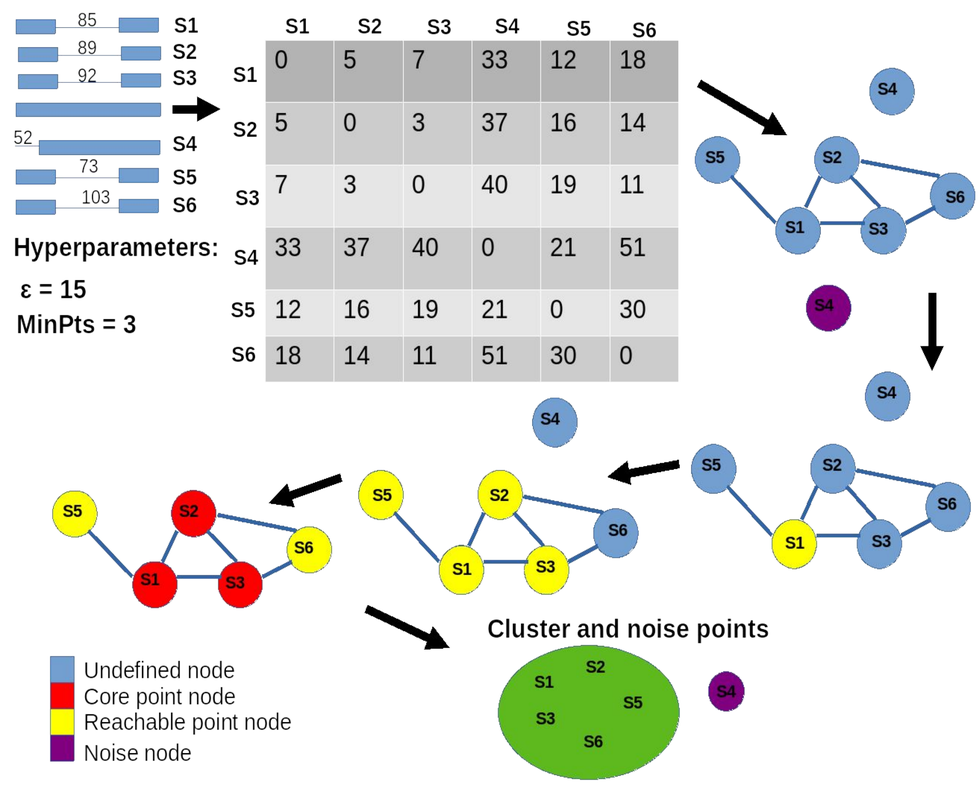
Signature collection

- Read alignments are processed in two ways: CIGAR string parsing and partial alignments analysis into signatures of deletions, insertions, and inversions.



Clustering phase: Distance matrix

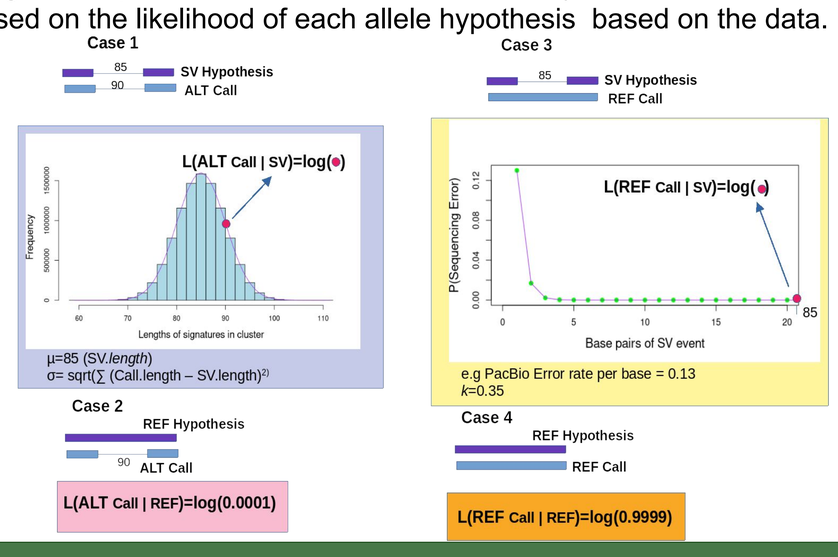
- A distance matrix is computed for all same type signatures from the distance between their reference coordinates and length.
- The DBSCAN algorithm is applied on the graph constructed from this matrix to cluster signatures, which then become SV calls.



	S1	S2	S3	S4	S5	S6
S1	0	5	7	33	12	18
S2	5	0	3	37	16	14
S3	7	3	0	40	19	11
S4	33	37	40	0	21	51
S5	12	16	19	21	0	30
S6	18	14	11	51	30	0

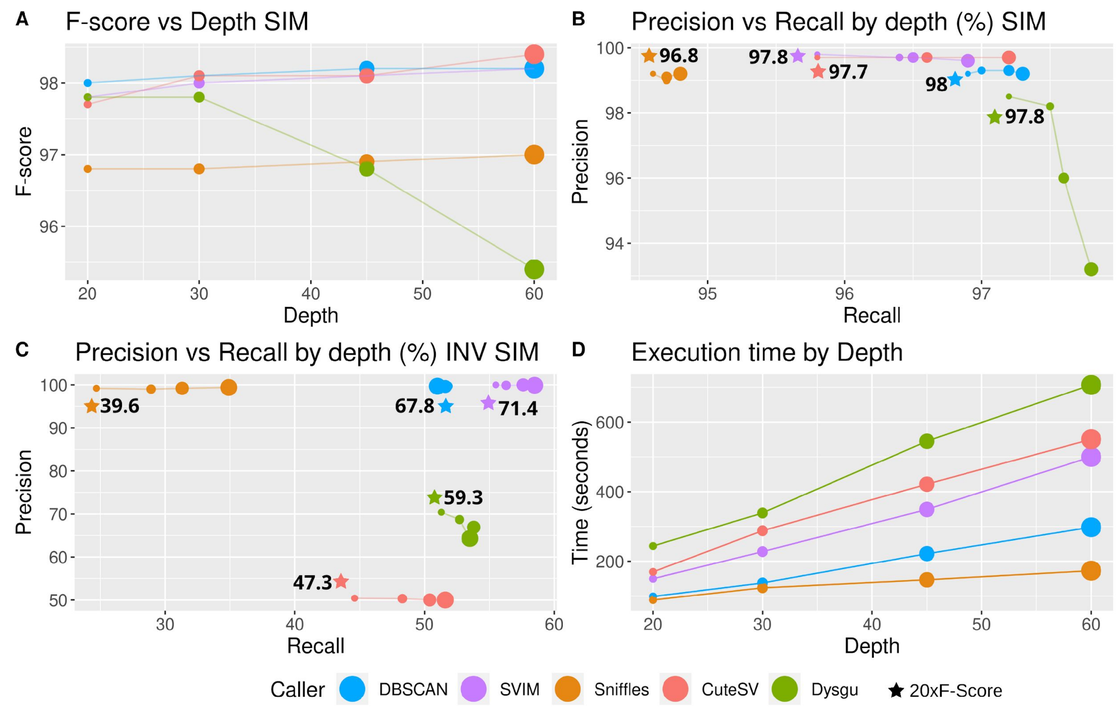
SV Genotyping

- Genotyping was implemented based on a previous bayesian model for SNPs (Gil et al., 2021), based on the likelihood of each allele hypothesis based on the data.



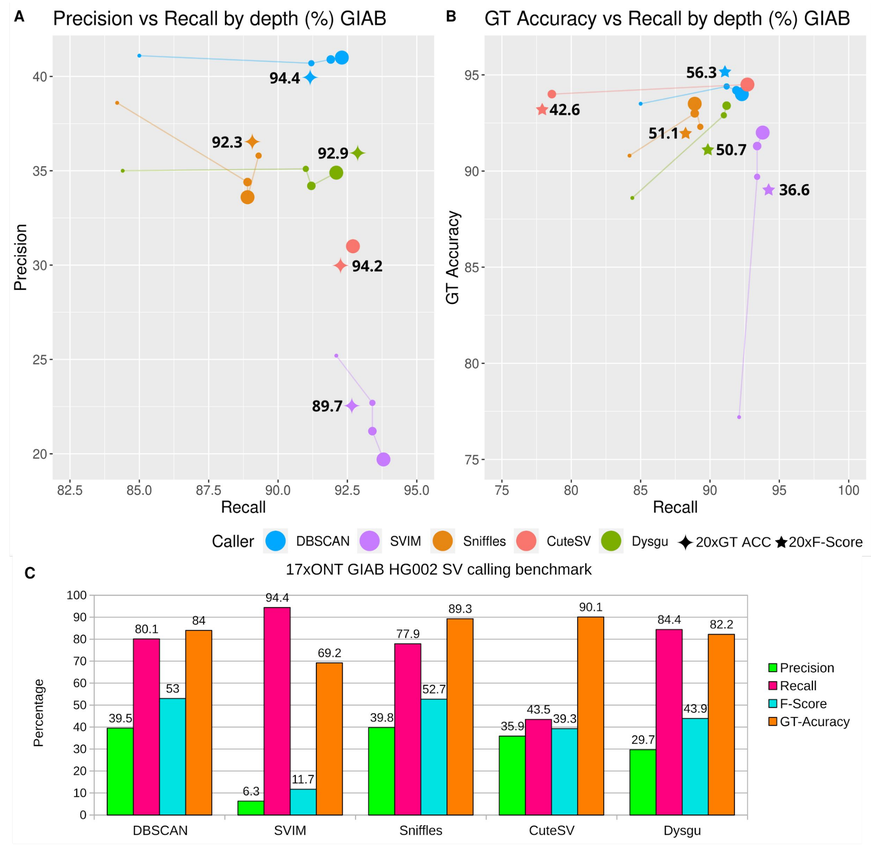
Benchmark: Simulation

- We performed SV indel and inversion simulation with VISOR on the TAIR10 genome.
- DBSCAN is able to produce the best f-score on all depths, including low depth datasets.
- Our tool was second best in single thread execution time.



Benchmark: GIAB dataset

- For real data we used the Gold Standard GIAB HG002 SV and read collections.
- DBSCAN outperformed the other tools in terms of precision, recall and genotyping accuracy of true positives with HiFi and ONT reads on the majority of depths.



Conclusions

- Benchmarking experiments indicate that our algorithm achieves superior accuracy for SV detection and genotyping compared to current software solutions.
- Even at 20x read depth, the integration of the Bayesian model provided the best results for genotyping accuracy, showing that our algorithm is robust to reductions of read depth.
- We expect that this new NGSEP functionality will be useful for a large number of ongoing and upcoming research in population genomics for different species.

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